KERALA SSLC | PHYSICS

ELECTRIC ENERGY

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

The resistance of a heating coil in a heating appliance operating at 200 V is 60 Ω.

(a) Which alloy is used to make heating coil of heating appliances ?

(b) Calculate the amount of heat produced if electricity flows through this appliance for 5 minutes.

(c) If another appliance with a resistance less than 60 Ω is operated at the same voltage for 5 minutes, will the amount of heat produced increase or decrease ? What is your justification ?

2. 2026 **2 MARKS**

(A) A heating appliance produces 7200000 J heat in 2 hours.

(a) What is the energy change in a heating appliance ?

(b) Calculate the power of this appliance.

OR

(B) An electric grinder with a power of 400 W is working.

(a) What is the energy change in this equipment ?

(b) Calculate the work done by this grinder in 1 minute.

3. 2026 **1 MARK**

What does the term "Ohmic Heating" indicate ?

(a) Heating in a microwave oven.

(b) Heating in an induction cooker.

(c) Joule heating.

(d) Heating through radiation.

4. 2025 **3 MARKS**

An electric heater has a resistance of 690 Ω and is designed to operate at 230 V.

(a) What are the characteristics of a good heating coil ?

(b) Calculate the heat energy produced by this heater when it works for 30 minutes.

5. 2024 **4 MARKS**

The resistance of an electrical device designed to work at 230 V is 460 Ω.

(a) Calculate the current drawn by this device when operated at 230 V.

(b) Find the rated power of the device.

(c) Calculate the heat produced if this device works for 10 minutes.

6. 2024 **2 MARKS**

Which material is used to make the filament of an incandescent lamp ? What properties make this material suitable for this purpose ?

7. 2023 **4 MARKS**

A heating coil with 60 Ω resistance is connected to a 240 V supply.

(a) Calculate the power of the appliance.

(b) Calculate the amount of heat generated by this heating coil in 5 minutes.

(c) If this appliance continuously works for 10 hours, calculate the energy consumed in commercial units.

8. 2023 **2 MARKS**

(a) What is the main disadvantage of an incandescent lamp ?

(b) Write any two properties of tungsten that make it suitable for being used as a filament.

9. 2022 **4 MARKS**

An incandescent lamp bears the marking 200 V, 100 W.

(a) What does 100 W indicate ?

(b) What is the resistance of its filament ?

(c) Write an advantage of LED lamp over incandescent lamp.

10. 2022 **1 MARK**

Which is the commercial unit of electrical energy ?

(ampere, kilowatt, kilowatt hour, volt)

11. 2022 **1 MARK**

Find the relation between the terms in the first pair and complete the second pair.

Incandescent lamp : Tungsten

Heating coil of heating appliances : _____

12. 2021 **3 MARKS**

A 2 A current flows through an electric heating device connected to 230 V supply.

(a) The quantity of charge that flows through the circuit in 5 minutes is :

(i) 10 C (ii) 60 C (iii) 600 C (iv) 6 C

(b) What is the resistance of the device ?

(c) Calculate the power of the heating device.

13. 2021 **2 MARKS**

(a) Name the part of a heating equipment in which the electric energy is converted into heat energy.

(b) Name the substance used to make this part.

14. 2021 **2 MARKS**

Write the energy change taking place in the given devices.

(a) Incandescent Lamp

(b) Electric Mixie